

Pharmacological Interventions for Weight Reduction in Patients With Schizophrenia Treated With Antipsychotics

A Systematic Review and Network Meta-Analysis

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IMPORTANCE Significant weight gain is a concerning adverse effect of antipsychotic medications experienced by patients with schizophrenia spectrum disorders (SSDs). Its high prevalence and significant contribution to cardiometabolic morbidity in this population warrant better consensus on the management of antipsychotic-induced weight gain and related comorbidity.

OBJECTIVES To evaluate the association between pharmacological interventions and changes in body weight among antipsychotic-treated patients with SSDs.

DATA SOURCES Ovid MEDLINE, Embase, PsycINFO, the Cochrane Central Register of Controlled Trials (CENTRAL), CINAHL, ClinicalTrials.gov, and the International Clinical Trials Registry Platform (ICTRP) Search Portal were searched up to December 5, 2025.

STUDY SELECTION Randomized clinical trials examining any pharmacological intervention for weight reduction in antipsychotic-treated patients with SSDs were included. No restrictions to study duration were applied.

DATA EXTRACTION AND SYNTHESIS A systematic review and frequentist random-effects network meta-analysis was conducted. Certainty in the evidence was assessed using the Confidence in Network Meta-Analysis (CINeMA) tool. The first round of data analysis took place between May 2025 to November 2025 and was updated in December 2025.

MAIN OUTCOMES AND MEASURES The primary outcome was change in body weight following treatment with pharmacological agent vs placebo or standard care. Secondary outcomes included other anthropometric and metabolic parameters.

RESULTS A total of 95 studies examining 39 individual pharmacological interventions were included in this review (pooled N = 5898). The network meta-analysis found that semaglutide (mean difference [MD], −10.98 kg; 95% CI, −13.33 to −8.62; k = 3; moderate certainty), liraglutide (MD, −5.43 kg; 95% CI, −8.54 to −2.33; k = 2; moderate certainty), topiramate (MD, −3.95 kg; 95% CI, −5.89 to −2.02; k = 5; moderate certainty), metformin (MD, −3.86 kg; 95% CI, −5.02 to −2.70; k = 16; moderate certainty), and exenatide (MD, −2.97 kg; 95% CI, −5.83 to −0.11; k = 3; moderate certainty) were associated with the most significant reductions in body weight compared to placebo. Other interventions including ramelteon, nizatidine, and aripiprazole were also found to be associated with weight-reducing effects but with very low certainty of evidence. Clinically meaningful weight change of 5% or greater was observed with semaglutide and metformin. Beneficial effects on other metabolic outcomes were also noted with several of the medications, and there were no major concerns with gastrointestinal adverse effects or leaving the study early (ie, dropouts) between interventions.

CONCLUSIONS AND RELEVANCE This systematic review and network meta-analysis found substantial variability in weight-related outcomes across pharmacological interventions for antipsychotic-treated individuals with SSDs. Semaglutide, liraglutide, topiramate, metformin, and exenatide were associated with the greatest reductions in body weight and were supported by the highest-certainty evidence, providing guidance for clinicians managing antipsychotic-associated weight gain.

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Antipsychotic medications are the cornerstone of treatment for schizophrenia spectrum disorders (SSDs).¹ Unfortunately, these agents compound already disproportionately high rates of cardiometabolic comorbidities, such as weight gain, glucose dysregulation, dyslipidemia, and type 2 diabetes, in a population already burdened by early cardiovascular mortality.²⁻⁴ These metabolic adverse effects also contribute to medication nonadherence, low self-esteem, decreased quality of life, and worse cognitive functioning,⁵⁻⁸ which collectively result in significantly greater health care costs.⁹ Thus, addressing the metabolic disturbances associated with antipsychotic use is necessary to improve the overall health and quality of life of individuals being treated with these medications.

Several systematic reviews and meta-analyses have examined the effectiveness of pharmacological and nonpharmacological interventions for attenuating antipsychotic-associated weight gain. While lifestyle and behavioral interventions have been extensively studied, their effect sizes are modest and may not translate to clinically significant outcomes (ie, $\geq 5\%$ or $\geq 7\%$ weight loss).^{10,11} Furthermore, adherence with lifestyle interventions is low among individuals with SSDs,^{12,13} undermining the cost-effectiveness and scalability.^{11,14} In this regard, adjunctive pharmacological interventions represent an important strategy to achieve clinically meaningful weight loss.

The use of pharmacological interventions to treat antipsychotic-associated weight gain has been previously synthesized, including in network meta-analyses from 2024 and 2025.¹⁵⁻¹⁸ Metformin consistently had the most evidence for its effectiveness in this population, followed by moderate evidence for other interventions, such as topiramate, nizatidine, and sibutramine. The field of neuroendocrinology has rapidly shifted in recent years, including introduction of newer glucagon-like peptide-1 receptor agonists (GLP-1 RAs) like semaglutide, which are revolutionizing management of obesity and related comorbidities. Existing reviews have included older GLP-1 RAs and focused on body weight reduction, excluding examination of other key cardiometabolic risk factors. Accordingly, the aim of this systematic review and network meta-analysis is to present a thorough and current synthesis of all available evidence on pharmacological interventions, including recently published trials on semaglutide, in relation to body weight and associated metabolic disturbances in individuals with SSDs receiving antipsychotic treatment. This is distinct from prevention of antipsychotic-associated weight gain (where metabolic agents are initiated concomitantly with antipsychotic treatment), which has been summarized in a different publication, given the difference in clinical contexts.¹⁹

Methods

This systematic review and network meta-analysis followed the reporting standard of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) extension statement for network meta-analysis.²⁰ The study protocol was registered in PROSPERO on November 15, 2024 (CRD42024610768;

Key Points

Question Which pharmacological interventions are associated with reducing body weight in individuals with schizophrenia spectrum disorders (SSDs) treated with antipsychotics?

Findings This systematic review and network meta-analysis, encompassing 95 studies and 39 distinct pharmacological interventions (pooled N = 5898), found that semaglutide, liraglutide, topiramate, metformin, and exenatide were associated with the most significant reductions in body weight compared to placebo. Beneficial effects on other metabolic outcomes were noted with several of the medications, and there were no major concerns with adverse effects or dropouts between interventions.

Meaning Adjunctive pharmacological interventions represent an important strategy to treat weight gain associated with antipsychotic use in SSDs.

deviations from the protocol are described in eAppendix 1 in Supplement 1).

Search Strategy and Inclusion Criteria

A systematic literature search was conducted from database inception to December 5, 2025, in Ovid MEDLINE, Ovid Embase, Ovid PsycINFO, the Cochrane Central Register of Controlled Trials (CENTRAL), CINAHL, ClinicalTrials.gov, and the International Clinical Trials Registry Platform (ICTRP) Search Portal; reference lists of identified studies and reviews were hand-searched for additional studies. The search consisted of terms for clinical trials of metabolic interventions in antipsychotic-treated patients with SSDs (eAppendix 2, eTable 1 in Supplement 1). Only randomized clinical trials (RCTs) conducted in adults (≥ 18 years old) were considered for inclusion. Studies focused on preventing antipsychotic-associated weight gain were not included in this review. There were no other restrictions based on illness duration, stage of illness, treatment setting, treatment duration, current clinical state, symptom cluster, or intervention dose. Data from non-peer-reviewed sources were included (eg, data published on ClinicalTrials.gov). Studies where more than 50% of the sample had a non-schizophrenia-related diagnosis were excluded. This criterion was applied to ensure that findings were primarily reflective of the schizophrenia population and to minimize confounding from studies in which outcomes may be driven by other diagnostic groups.²¹

Study Selection

Study screening and selection was conducted using Coviidence. Article screening at both the title/abstract and full-text levels was completed by a team of 4 authors (N.S., S.M.A., K.M., M.K.H.), with 2 authors independently screening articles at each stage. Any disagreements were resolved by discussion among all authors.

Data Extraction and Processing

All data were extracted using standardized data extraction forms by a team of 4 authors (N.S., S.M.A., K.M., H.F.), with

each study independently reviewed by 2 authors. The primary outcome was change in body weight following treatment with the pharmacological agent vs placebo or standard care. Secondary outcomes included other anthropometric or metabolic parameters, such as clinically meaningful weight loss (ie, $\geq 5\%$ and $\geq 7\%$ weight loss), body mass index (BMI, calculated as weight in kilograms divided by height in meters squared), blood pressure, waist circumference, fasting glucose, fasting insulin, hemoglobin A_{1c}, homeostatic model assessment for insulin resistance (HOMA-IR), total cholesterol, high-density lipoprotein cholesterol, low-density lipoprotein cholesterol, and triglycerides, as well as feasibility and tolerability measures, such as the number of individuals leaving the study early for any reason (also referred to as dropouts) or experiencing gastrointestinal adverse effects, such as nausea, diarrhea, or vomiting. Continuous data were extracted as mean and SD. Dichotomous data were extracted as number of events that occurred in the total sample. Corresponding authors were contacted if data were unavailable or could not be extracted in a usable form from the published article. More details on data extraction and processing can be found in eAppendix 2 in Supplement 1.

Risk of Bias Assessment

The Cochrane Risk of Bias 2 tool was used to assess bias in individual trials for the primary outcome (body weight) and was conducted by 2 independent reviewers (N.S., S.M.A., K.M.). Any conflicts were resolved by group discussion. Studies were considered high risk of bias if at least 1 of the domains was labeled as high risk.

Statistical Analysis

Assessment of the Transitivity Assumption

To ensure transitivity in the network,²² analyses were restricted to patients with SSDs. Visual inspection of other effect modifiers (eg, age, sex, baseline BMI, illness duration, years of antipsychotic treatment exposure, intervention duration, and percentage White) was done to ensure a network meta-analysis could be conducted.

Network Meta-Analysis and Ranking of Interventions

All analyses were done in R version 4.2.3 (R Foundation). A frequentist random-effects network meta-analysis of all reported interventions was conducted using the *netmeta* package (version 3.2-0). A random-effects model was chosen to reflect possible methodological differences across trials (eg, in participants, interventions, or definitions of outcomes). Continuous outcome variables (ie, all anthropometric and metabolic parameters) were estimated using mean difference (MD) and 95% confidence intervals; dichotomous outcomes (ie, clinically meaningful weight loss, adverse effects, and number of dropouts) were estimated using odds ratios (ORs) and 95% confidence intervals. Placebo or usual care was used as the reference treatment in all forest plots and league tables. A network plot for each outcome was constructed using the *netgraph* function. Custom forest plots were constructed using *ggplot2*, where interventions are arranged based on descending MD and interventions with a significant effect are presented

before those found to be insignificant. *P*-scores were used to rank metabolic interventions based on their relative performance across outcomes. Studies not included in the meta-analysis (ie, those with no available extractable data or of interventions that were withdrawn from the market) were summarized narratively. Two-tailed *P* < .05 was considered significant.

Heterogeneity and Inconsistency Assessment

Heterogeneity was assessed using τ and I^2 statistics, with higher values indicating greater variability in treatment effects.²³ Network consistency was assessed using both global (design-by-treatment interaction model and Cochran Q statistic) and local approaches (separate indirect from direct evidence [SIDE] method).²⁴ Prediction intervals (PIs) showing the expected range of treatment effects in future studies are presented in all forest plots (unless there were <3 studies included, in which case PIs could not be calculated).

Assessment of the Certainty of Evidence

Certainty of evidence for the primary outcome (body weight) was rated as high, moderate, low, or very low using the Confidence in Network Meta-Analysis (CINeMA) application (eAppendix 2.4 in Supplement 1), and this rating was used to color-code interventions in the forest plot.

Sensitivity Analyses

Sensitivity analyses were conducted for the primary body weight outcome by removing studies based on a specific rule (ie, removing agents that have been withdrawn from the market or studies with high risk of bias, end point data only, those translated to English from another language using Google Translate, or non-peer-reviewed data coming from clinical trial websites).

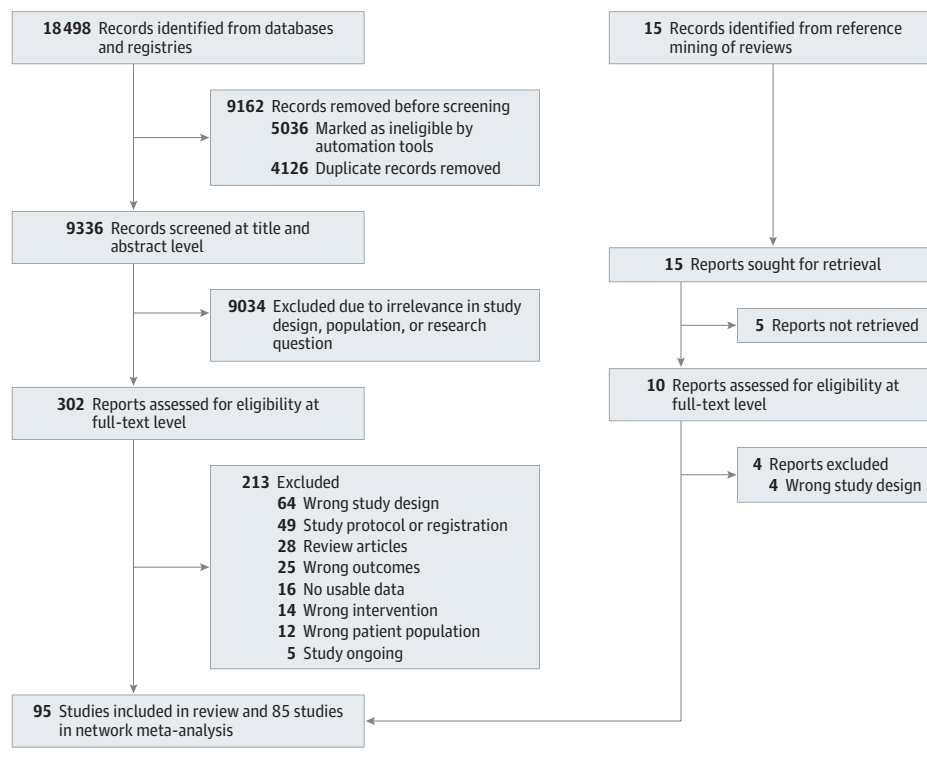
Results

Characteristics of Included Studies

The initial search identified 18 498 results. Following removal of duplicates and title and abstract screening, 302 studies were assessed for full-text eligibility; an additional 10 studies were reviewed from reference mining. A total of 95 studies, published between 1965 and 2025, met inclusion criteria and were included in the final review; 85 studies were synthesized in the quantitative analysis (Figure 1).

The 95 trials (references in eAppendix 2.6 in Supplement 1) included in this review reported on the following 39 independent medications: metformin (*k* = 20); aripiprazole (*k* = 11); topiramate (*k* = 8); sibutramine (*k* = 4); exenatide, semaglutide, nizatidine, betahistine, and ω -3 fatty acids (*k* = 3 each); liraglutide, fluvoxamine, rosiglitazone, ranitidine, orlistat, naltrexone + bupropion, naltrexone, amantadine, and melatonin (*k* = 2 each); and metformin + sibutramine, metformin + lorcaserin, fluoxetine, dextroamphetamine, modafinil, D-fenfluramine, phenylpropanolamine, atomoxetine, pioglitazone, intranasal insulin, sitagliptin, pramlintide, lorcaserin, rimonabant, telmisartan, rosuvasta-

Figure 1. PRISMA Flowchart of Included Studies



tin, pravastatin, zonisamide, ramelteon, minocycline, and celecoxib ($k = 1$ each). Characteristics of each individual study, organized by metabolic intervention, can be found in eTable 2 in Supplement 1. A brief overview of each drug class and its likely mechanism of action is provided in eAppendix 3.1 in Supplement 1. Ten studies were reported on narratively in eAppendix 3.3 in Supplement 1 (3 had no usable data, while 7 were of interventions withdrawn from the market).

No major violations in the transitivity assumption were detected on comparison of clinical characteristics of studies across comparisons (described in more detail in eAppendix 3.2, eFigures 1 and 2 in Supplement 1). For the body weight outcome, there was a single closed loop involving placebo, metformin, and topiramate (Figure 2); all other outcomes formed a purely star pattern network (eFigure 2 in Supplement 1). This indicates that most evidence comes from direct comparisons with the common comparator (placebo).

Risk of Bias Evaluation

Forty studies were deemed to be of high risk of bias (eTable 4 in Supplement 1). Sensitivity analyses for the primary outcome were conducted excluding these studies to evaluate their impact on the overall effect within their respective comparisons (eAppendix 3.5 in Supplement 1).

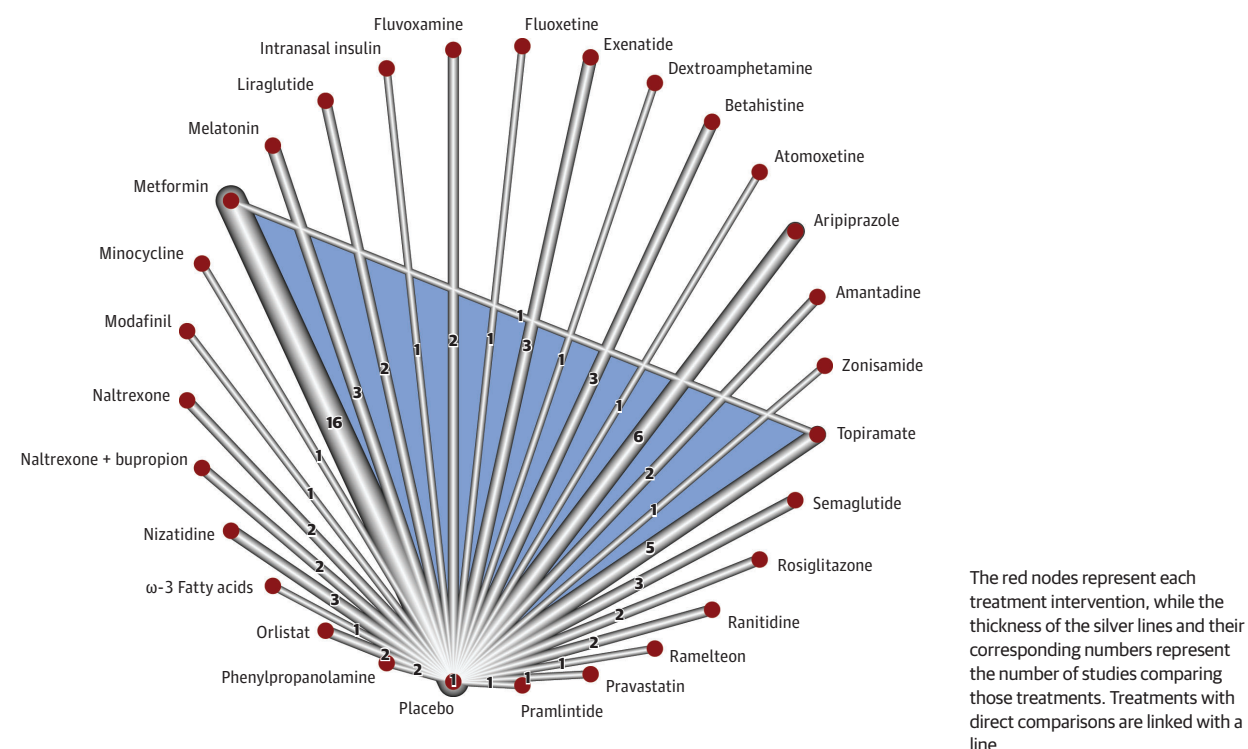
Continuous Outcomes

Body Weight

A total of 69 trials studied the effect of 28 interventions on body weight. A reduction in body weight was observed with sema-

glutide (MD, -10.98 kg; 95% CI, -13.33 to -8.62 ; $k = 3$; moderate certainty), ramelteon (MD, -9.00 kg; 95% CI, -17.78 to -0.22 ; $k = 1$; very low certainty), liraglutide (MD, -5.43 kg; 95% CI, -8.54 to -2.33 ; $k = 2$; moderate certainty), nizatidine (MD, -4.46 kg; 95% CI, -7.94 to -0.98 ; $k = 3$; very low certainty), topiramate (MD, -3.95 kg; 95% CI, -5.89 to -2.02 ; $k = 5$; moderate certainty), metformin (MD, -3.86 kg; 95% CI, -5.02 to -2.70 ; $k = 16$; moderate certainty), exenatide (MD, -2.97 kg; 95% CI, -5.83 to -0.11 ; $k = 3$; moderate certainty), and aripiprazole (MD, -2.37 kg; 95% CI, -4.09 to -0.64 ; $k = 6$; very low certainty) compared with placebo (Figure 3; CINeMA ratings presented in eTable 5 in Supplement 1; league table presented in eTable 23 in Supplement 2). The overall network for this outcome demonstrated substantial heterogeneity (τ , 1.67 kg; I^2 , 68.4%). Global tests indicated significant inconsistency within the network (Q , 132.99; $P < .001$), which was further supported by local inconsistency identified via the back-calculation method, particularly for the metformin vs placebo comparison (eAppendix 3.10, eTable 21 in Supplement 1). P -scores identified the most efficacious interventions to be semaglutide, ramelteon, liraglutide, nizatidine, topiramate, and metformin (eAppendix 3.9, eTable 6 in Supplement 1). Ramelteon and nizatidine were no longer included as comparisons when studies with high risk of bias were excluded. Semaglutide, liraglutide, topiramate, exenatide, and metformin were consistently found to be the most efficacious agents across all other sensitivity analyses (eAppendix 3.5 in Supplement 1). A subgroup analysis based on intervention duration (<24 weeks or ≥ 24 weeks) revealed that ramelteon, liraglutide, nizatidine,

Figure 2. Network Plot of All Studies Included in the Body Weight Outcome Comparison



metformin, topiramate, and aripiprazole were most efficacious among the short-term studies, while semaglutide and metformin were the most efficacious among the long-term studies (eAppendix 3.6, eFigure 7 in [Supplement 1](#)). In a subgroup analysis of studies including only clozapine-treated patients, semaglutide was the most efficacious intervention, followed by exenatide, metformin, ω-3 fatty acids, and aripiprazole (eAppendix 3.6, eFigure 8A in [Supplement 1](#)). In contrast, no interventions reached statistical significance in the olanzapine-only studies (eAppendix 3.6, eFigure 8B in [Supplement 1](#)). However, there were a limited number of studies per intervention (1-3 studies each) in these subgroup analyses by primary antipsychotic treatment.

The network meta-analyses for other weight-related measures (ie, BMI and waist circumference) and metabolic parameters are reported in eAppendix 3.7 in [Supplement 1](#).

Dichotomous Outcomes

Clinically Significant Weight Loss (≥5%)

Eight trials reported on the number of individuals that achieved clinically significant weight loss of 5% or greater with long-term treatment across 5 interventions. This outcome was achieved with semaglutide (OR, 210.00; 95% CI, 3.97-11115.99; $k = 1$) and metformin (OR, 10.39; 95% CI, 1.89-59.54; $k = 4$) but not with any of the other interventions compared to placebo ([Figure 4A](#)). The overall network for this outcome demonstrated moderate heterogeneity and inconsistency (τ , 1.40; I^2 , 65.2%). Global tests indicated significant inconsistency within the network (Q , 8.62; $P = .03$); however, no closed loops were present in this net-

work, which precluded local inconsistency from being evaluated.

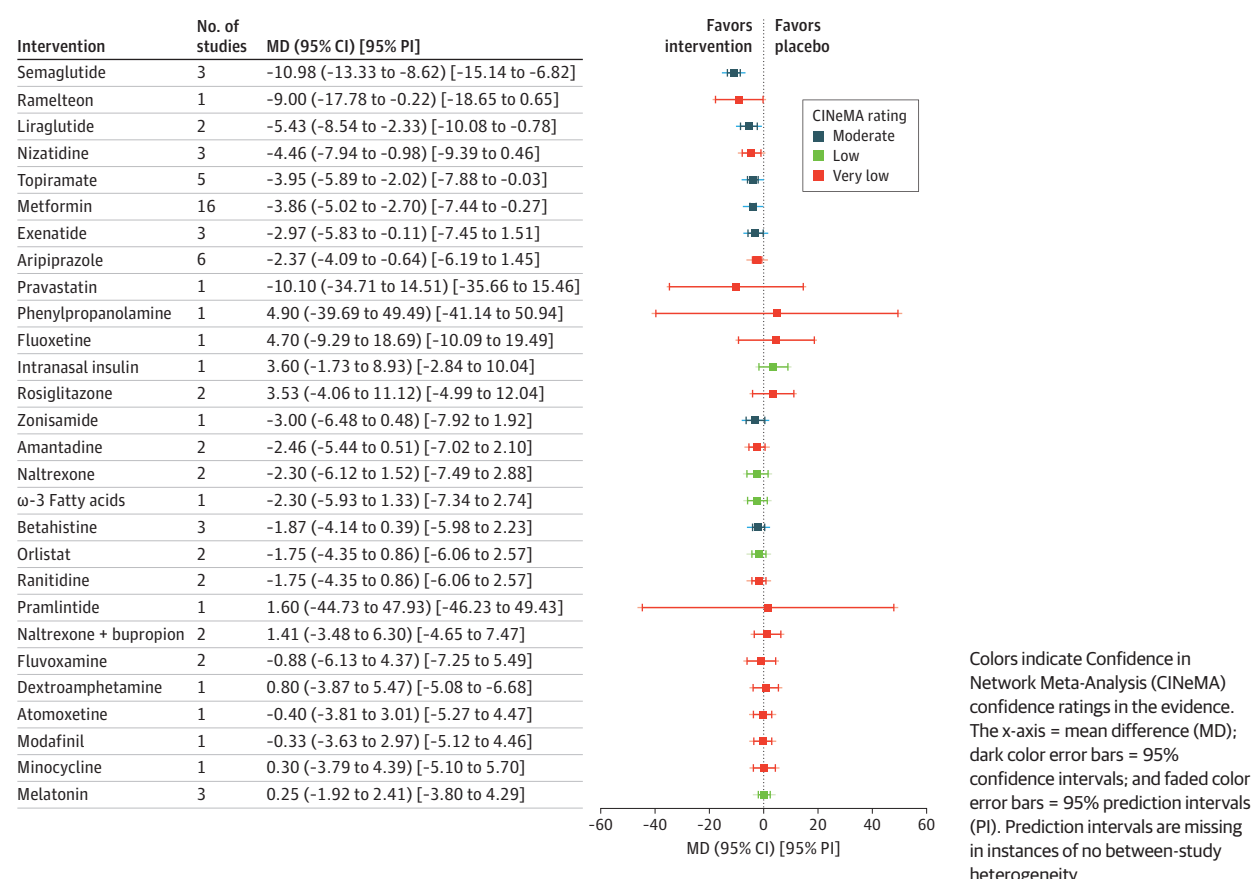
Number of Participants Leaving the Study Early for Any Reason (Study Dropouts)

A total of 53 studies recorded the number of participants dropping out prematurely (for any reason) across 28 interventions. There were no differences between any of the treatments and placebo for this outcome ([Figure 4B](#)). The frequency of gastrointestinal adverse effects (vomiting, nausea, and diarrhea) is reported in eAppendix 3.7 in [Supplement 1](#).

Discussion

This systematic review and network meta-analysis, encompassing 95 RCTs of 39 unique interventions, highlights differences across pharmacological agents in weight and metabolic outcomes associated with antipsychotic treatment relative to both placebo and one another. Specifically, semaglutide, liraglutide, exenatide, metformin, and topiramate showed weight loss effects ranging from 2.96 to 10.97 kg with moderate certainty of evidence. Although aripiprazole, ramelteon, and nizatidine similarly demonstrated some weight-reducing effects, the certainty of evidence for these comparisons was very low, precluding firm conclusions from being made. Beneficial effects of these agents on other metabolic outcomes, such as lipid and glucose levels, were also noted, supporting further investigation. Overall, the

Figure 3. Forest Plot Showing the Effect of Pharmacological Interventions on Body Weight



medications were found to have a favorable tolerability profile, with equal attrition rates between the active and placebo arms.

Treatment with semaglutide was associated with a pronounced weight loss of almost 11 kg, translating to clinically meaningful weight loss ($\geq 5\%$ baseline weight) and exceeding that observed with any other pharmacological interventions in the network. These trends align with findings from network meta-analyses conducted in the general population, which showed an 11% reduction in body weight.²⁵ This suggests that the weight-lowering potential of semaglutide observed in general obesity trials may extend to individuals treated with antipsychotics. This is in contrast to other weight loss drugs approved for chronic treatment of obesity, such as orlistat and naltrexone-bupropion, that have not demonstrated similar effectiveness for people treated with antipsychotics, suggesting that the underlying mechanisms of weight gain differ in the context of antipsychotic treatment, warranting trials to ensure safety and efficacy in this population.

In this review, topiramate on its own was used off-label and provided moderate evidence of reducing antipsychotic-associated weight gain; there were no studies of phentermine-topiramate, as it is seldomly used due to concerns of triggering a relapse of psychosis or mania.²⁶ Meanwhile, naltrexone-bupropion did not produce signifi-

cant weight loss in our network analysis. As such, newer weight loss agents, such as semaglutide, may offer greater weight loss potential in this specific population. Regardless, interpretation should remain cautious given the relatively small evidence base. With respect to metformin, our results are largely consistent with those of previous network meta-analyses,¹⁵⁻¹⁸ with metformin remaining the most thoroughly investigated agent for addressing antipsychotic-associated weight gain, accounting for 22% of the studies included in this review.

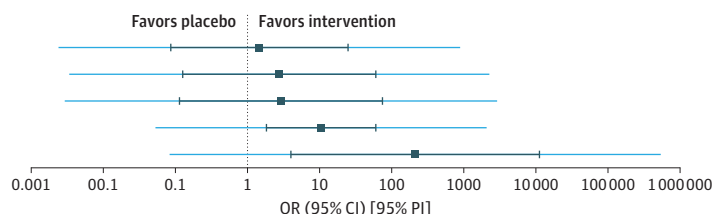
Strengths and Limitations

A strength of this study is that it is founded on diverse data that lend to the generalizability of our findings, summarizing evidence from both inpatient and outpatient settings, across several geographical continents, and among patients of varying disease severity. Furthermore, to be as comprehensive as possible, we included data from non-peer-reviewed sources, such as the ClinicalTrials.gov registry, for terminated or unpublished studies. As this may introduce biases, we conducted sensitivity analyses removing these individual studies to ensure data quality in the final synthesis.

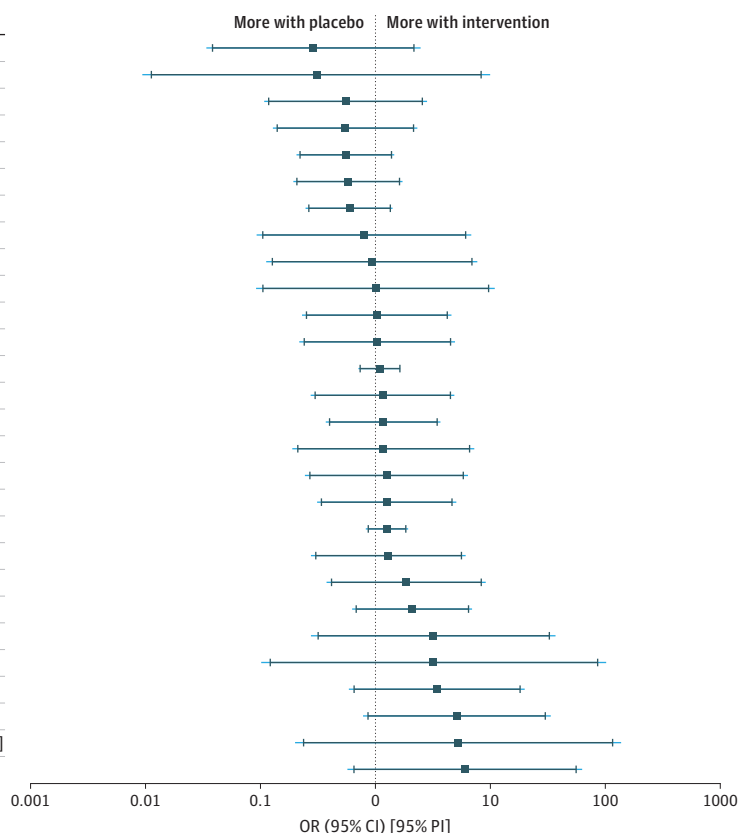
To our knowledge, this synthesis offers the most extensive and current assessment of the literature to date, including evidence from newer GLP-1 RAs, such as semaglutide, while

Figure 4. Forest Plots Showing the Effect of Pharmacological Interventions on Dichotomous Outcomes: Clinically Meaningful Weight Loss ($\geq 5\%$, A) and Leaving the Study Early for Any Reason (B)**A** Clinically meaningful weight loss

Intervention	No. of studies	OR (95% CI) [95% PI]
Aripiprazole	1	1.45 (0.09-24.59) [0-876.06]
Amantadine	1	2.74 (0.13-59.75) [0-2232.16]
Orlistat	1	2.88 (0.11-73.73) [0-2854.87]
Metformin	4	10.39 (1.81-59.54) [0.05-2047.28]
Semaglutide	1	210.00 (3.97-11 115.99) [0.08-531 616.45]

**B** Leaving the study early (for any reason)

Intervention	No. of studies	OR (95% CI) [95% PI]
Ramelteon	1	0.29 (0.04-2.17) [0.03-2.41]
Naltrexone + bupropion	1	0.30 (0.01-8.32) [0.01-9.84]
Fluoxetine	1	0.55 (0.12-2.55) [0.11-2.76]
Telmisartan	1	0.55 (0.14-2.14) [0.13-2.29]
ω -3 Fatty acids	2	0.55 (0.22-1.37) [0.21-1.43]
Topiramate	3	0.58 (0.21-1.61) [0.20-1.70]
Semaglutide	3	0.59 (0.26-1.34) [0.25-1.39]
Pioglitazone	1	0.79 (0.10-6.07) [0.09-6.73]
Nizatidine	2	0.93 (0.13-6.85) [0.11-7.59]
Phenylpropanolamine	1	1.00 (0.10-9.61) [0.09-10.78]
Atomoxetine	1	1.03 (0.25-4.24) [0.23-4.55]
Orlistat	1	1.03 (0.24-4.50) [0.22-4.84]
Aripiprazole	5	1.10 (0.74-1.63) [0.72-1.66]
Betahistine	1	1.15 (0.30-4.47) [0.28-4.78]
Sitagliptin	1	1.16 (0.40-3.41) [0.38-3.60]
Intranasal insulin	1	1.17 (0.21-6.51) [0.19-7.11]
Pramlintide	1	1.25 (0.27-5.83) [0.25-6.30]
Pravastatin	1	1.25 (0.34-4.64) [0.31-4.96]
Metformin	14	1.25 (0.86-1.83) [0.84-1.87]
Naltrexone	2	1.30 (0.30-5.61) [0.28-6.04]
Exenatide	3	1.85 (0.41-8.30) [0.38-8.96]
Liraglutide	2	2.09 (0.67-6.49) [0.64-6.88]
Rosuvastatin	1	3.21 (0.32-32.60) [0.28-36.58]
Rosiglitazone	1	3.21 (0.12-85.20) [0.10-100.65]
Fluvoxamine	1	3.42 (0.65-18.00) [0.60-19.59]
Modafinil	1	5.10 (0.86-30.27) [0.79-33.13]
Zonisamide	1	5.26 (0.24-116.57) [0.20-136.44]
Melatonin	2	6.00 (0.65-55.66) [0.58-62.33]



The x-axis (odds ratio [OR]) scales differ between panels. Dark color error bars = 95% confidence intervals; faded color error bars = 95% prediction intervals (PI). Prediction intervals are missing in instances of no between-study heterogeneity.

also excluding compounds withdrawn from the market to ensure the findings are clinically relevant. Furthermore, the network approach allowed indirect comparisons between treatments that have not been evaluated head to head, providing a more comprehensive synthesis of the available evidence than pairwise meta-analyses alone. These results help clarify the relative positioning of commonly used and emerging pharmacological strategies for antipsychotic-associated weight gain.

Despite the comprehensive nature of the current review, several limitations must be acknowledged. First, the shape of the networks for most outcomes in this review was primarily

star shaped, with only a single closed loop between metformin, topiramate, and placebo. Since few direct head-to-head trials compared active interventions, this network meta-analysis primarily relied on indirect evidence via a single common comparator, which may lead to less precise effect estimates and wider confidence intervals.²⁷ Additionally, star-shaped networks do not support inconsistency testing, which can limit the certainty of the results. Therefore, the *P*-score rankings should be interpreted with great caution. While we explored the assumption of transitivity by examining the distribution of potential effect modifiers across treatment comparisons and ensuring clinical and methodological similarity,

there remains an inherent risk of violating this assumption. Furthermore, most of the individual interventions had only 1 to 2 studies, which further reduces confidence in the estimates and limits generalizability. There was also considerable heterogeneity in our networks, which may reflect differences in intervention dose, study duration or follow-up periods, and other key methodological factors across the included studies. To explore potential sources of heterogeneity, we conducted a subgroup analysis according to intervention duration and several sensitivity analyses based on different rules, such as excluding non-English studies; however, the heterogeneity and inconsistency of the networks remained high in these additional analyses. The state of evidence on this topic could be considerably improved by head-to-head trials.

In terms of the reported outcomes, not all studies looked at metabolic end points beyond weight change, which prevented proper evaluation of these interventions on key non-weight-related outcomes. In addition to reporting absolute weight change, future studies should report on percentage weight change and the proportion of individuals achieving clinically meaningful weight loss, as well as functional and quality of life outcomes, to provide a more comprehensive assessment of clinical impact. Adverse effects and tolerability were also not evaluated systematically using structured scales, which made it difficult to extract this information; however, leaving the study early was used as a proxy outcome to address this. Lastly, less than one-fifth of the included studies were long term, limiting the evaluation of the interventions' extended efficacy and safety.

Conclusions

To conclude, this network meta-analysis indicates that semaglutide, liraglutide, exenatide, metformin, and topiramate were associated with the greatest effects in treating antipsychotic-associated weight gain in people with SSDs, with metformin remaining the most studied option for this purpose. The quality rankings for all these interventions were moderate, highlighting the need for future RCTs that are adequately powered and have a longer treatment duration to evaluate efficacy and safety. Given issues of affordability and access to newer interventions, such as semaglutide, sequenced treatment interventions can help address questions of equity and accessibility. Moreover, it may also be worthwhile to research newer GLP-1 RAs, such as tirzepatide, which have shown potential for significant weight loss and cardiovascular risk reduction in the general population. Regardless, given the prevalence of antipsychotic-associated weight gain and the significant impact on morbidity and well-being beyond cardiometabolic health, clinicians must remain vigilant with metabolic monitoring practices and intervene at the earliest stage possible. With the rapidly evolving field of obesity management, there is potential for patients with SSDs to benefit from current and emerging treatments across research and clinical contexts; however, deliberate efforts, including the development and update of clinical guidelines to incorporate GLP-1 RA evidence, are required to support their implementation in public sector settings to ensure equitable access and avoid perpetuating disparities in care.

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